

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

05

10 answers · Rationale-rich · Teach from doc

Q1**B****B. a**

Choice B is correct. A cube has 6 faces of equal area, so if the total surface area of a cube is $6\left(\frac{a}{4}\right)^2$, then the area of one face is $\left(\frac{a}{4}\right)^2$. Likewise, the area of one face of a cube is the

square of one of its edges; therefore, if the area of one face is $\left(\frac{a}{4}\right)^2$, then the length of one edge of the cube is $\frac{a}{4}$. Since the perimeter of one face of a cube is four times the length of one edge, the perimeter is $4\left(\frac{a}{4}\right) = a$.

Choice A is incorrect because if the perimeter of one face of the cube is $\frac{a}{4}$, then the total surface area of the cube is $6\left(\frac{a}{4}\right)^2 = 6\left(\frac{a}{16}\right)^2$, which is not $6\left(\frac{a}{4}\right)^2$. Choice C is incorrect

because if the perimeter of one face of the cube is $4a$, then the total surface area of the cube is $6\left(\frac{4a}{4}\right)^2 = 6a^2$, which is not $6\left(\frac{a}{4}\right)^2$. Choice D is incorrect because if the perimeter of one face

of the cube is $6a$, then the total surface area of the cube is $6\left(\frac{6a}{4}\right)^2 = 6\left(\frac{3a}{2}\right)^2$, which is not $6\left(\frac{a}{4}\right)^2$.

Q2**-12**

The correct answer is -12. It's given that function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. It follows that the graph of $y = f(x)$ in the xy -plane passes through the points $(-20, 0)$ and $(-4, 0)$. When the graph of a quadratic function contains two points $(a, 0)$ and $(b, 0)$, the x -coordinate of the vertex of the graph is the average of a and b . Therefore, the x -coordinate of the vertex of the graph of $y = f(x)$ is $\frac{-20 + -4}{2}$, or -12. It's given that the graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. It follows that the value of r is -12.

Q3**25/4**

Accept also: 6.25

The correct answer is $\frac{25}{4}$. The given equation can be rewritten in the form $fx = ax - h^2 + k$, where a , h , and k are constants. When $a > 0$, h is the value of x for which fx reaches its minimum. The given equation can be rewritten as $fx = 4x^2 - \frac{50}{4}x + 126$, which is equivalent to $fx = 4x^2 - \frac{50}{4}x + \frac{50^2}{8} - \frac{50^2}{8} + 126$. This equation can be rewritten as $fx = 4x - \frac{50^2}{8} - \frac{50^2}{8} + 126$, or $fx = 4x - \frac{50^2}{8} - 4\frac{50^2}{8} + 126$, which is equivalent to $fx = 4x - \frac{25^2}{4} - \frac{121}{4}$. Therefore, $h = \frac{25}{4}$, so the value of x for which fx reaches its minimum is $\frac{25}{4}$. Note that 25/4 and 6.25 are examples of ways to enter a correct answer.

Q4**25**

The correct answer is 25. The value of $g(7 - w)$ is the value of gx when $x = 7 - w$, where w is a constant. Substituting $7 - w$ for x in the given equation yields $g(7 - w) = 7 - w(7 - w) - 27 - w + 6^2$, which is equivalent to $g(7 - w) = 7 - w(5 - w)(13 - w^2)$. It's given that the value of $g(7 - w)$ is 0.

Substituting 0 for $g(7 - w)$ in the equation $g(7 - w) = 7 - w(5 - w)(13 - w^2)$ yields $0 = 7 - w(5 - w)(13 - w^2)$. Since the product of the three factors on the right-hand side of this equation is equal to 0, at least one of these three factors must be equal to 0. Therefore, the possible values of w can be found by setting each factor equal to 0. Setting the first factor equal to 0 yields $7 - w = 0$. Adding w to both sides of this equation yields $7 = w$. Therefore, 7 is one possible value of w . Setting the second factor equal to 0 yields $5 - w = 0$. Adding w to both sides of this equation yields $5 = w$. Therefore, 5 is a second possible value of w . Setting the third factor equal to 0 yields $13 - w^2 = 0$. Taking the square root of both sides of this equation yields $13 - w = 0$. Adding w to both sides of this equation yields $13 = w$. Therefore, 13 is a third possible value of w . Adding the three possible values of w yields $7 + 5 + 13$, or 25. Therefore, the sum of all possible values of w is 25.

Q5**B****B. $-3 < x < 1$**

Choice B is correct. The graph of a quadratic function in the xy -plane is a parabola. The axis of symmetry of the parabola passes through the vertex of the parabola. Therefore, the vertex of the parabola and the midpoint of the segment between the two x -intercepts of the graph have the same x -coordinate. Since $f(-3) = f(-1) = 0$, the x -coordinate of the vertex is

$\frac{(-3) + (-1)}{2} = -2$. Of the shown intervals, only the interval in choice B contains -2 . Choices

A, C, and D are incorrect and may result from either calculation errors or misidentification of the graph's x -intercepts.

Q6**A**

A. $y = 331,776^{x+1}$

Choice A is correct. The y -intercept of a graph in the xy -plane is the point where $x = 0$. Substituting 0 for x in the given equation, $y = 576^{2x+2}$, yields $y = 576^{2(0)+2}$, which is equivalent to $y = 576^2$, or $y = 331,776$. Therefore, the graph of the given equation in the xy -plane has a y -intercept of 0, 331,776. It follows that $r = 0$ and $s = 331,776$. Thus, the equivalent equation $y = 331,776^{x+1}$ displays the value of s as the base.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**D****D. Neither I nor II**

Choice D is correct. A y -intercept of a graph in the xy -plane is a point where the graph intersects the y -axis, or a point where $x = 0$. Substituting 0 for x in the equation defining function f yields $f(0) = a2 \cdot 2^0 + 2 \cdot 2^b$, or $f(0) = a + 2 \cdot 2^b$. So, the y -coordinate of the y -intercept of the graph is $a + 2 \cdot 2^b$, or equivalently, $a + a2 \cdot 2^b$. It's given that function g is equivalent to function f , where $0 < a < b$. It follows that $k = 2 \cdot 2^b$. Since $a2 \cdot 2^b$ can't be equal to 0, the coefficient a can't be equal to $a + a2 \cdot 2^b$. Since $0 < a$, the constant k , which is equal to $2 \cdot 2^b$, can't be equal to $a + a2 \cdot 2^b$. Therefore, function g doesn't display the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane as a constant or coefficient. It's also given that function h is equivalent to function f , where $0 < a < b$. The equation defining f can be rewritten as $fx = a2 \cdot 2^x + a2 \cdot 2^b$. It follows that $m = a2 \cdot 2^b$. Since $a2 \cdot 2^b$ can't be equal to 0, the coefficient a can't be equal to $a + a2 \cdot 2^b$. Since $0 < a$, the constant m , which is equal to $a2 \cdot 2^b$, can't be equal to $a + a2 \cdot 2^b$. Therefore, function h doesn't display the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane as a constant or coefficient. Thus, neither function g nor function h displays the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane as a constant or coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q8**D**

D. $h = -16t - 1.8^2 + 51.84$

Choice D is correct. An equation representing the height above ground h , in meters, of a softball t seconds after it is launched by a machine from ground level can be written in the form $h = -at - b^2 + c$, where a , b , and c are positive constants. In this equation, b represents the time, in seconds, at which the softball reaches its maximum height of c meters above the ground. It's given that this softball reaches a maximum height of 51.84 meters above the ground at 1.8 seconds; therefore, $b = 1.8$ and $c = 51.84$. Substituting 1.8 for b and 51.84 for c in the equation $h = -at - b^2 + c$ yields $h = -at - 1.8^2 + 51.84$. It's also given that this softball hits the ground at 3.6 seconds; therefore, $h = 0$ when $t = 3.6$. Substituting 0 for h and 3.6 for t in the equation $h = -at - 1.8^2 + 51.84$ yields $0 = -a(3.6) - 1.8^2 + 51.84$, which is equivalent to $0 = -3.24a + 51.84$. Adding $3.24a$ to both sides of this equation yields $3.24a = 51.84$. Dividing both sides of this equation by 3.24 yields $a = 16$. Substituting 16 for a in the equation $h = -at - 1.8^2 + 51.84$ yields $h = -16t - 1.8^2 + 51.84$. Therefore, $h = -16t - 1.8^2 + 51.84$ represents the height above ground h , in meters, of this softball t seconds after it is launched.

Choice A is incorrect. This equation represents a situation where the maximum height is 3.6 meters above the ground at 0 seconds, not 51.84 meters above the ground at 1.8 seconds.

Choice B is incorrect. This equation represents a situation where the maximum height is 51.84 meters above the ground at 0 seconds, not 1.8 seconds.

Choice C is incorrect and may result from conceptual or calculation errors.

Q9**5**

The correct answer is 5. Let x represent the width, in inches, of the rectangle. It's given that the length of the rectangle is 4 inches less than 7 times its width, or $7x - 4$ inches. The area of a rectangle is equal to its width multiplied by its length. Multiplying the width, x inches, by the length, $7x - 4$ inches, yields $x(7x - 4)$ square inches. It's given that the rectangle has an area of 155 square inches, so it follows that $x(7x - 4) = 155$, or $7x^2 - 4x = 155$. Subtracting 155 from both sides of this equation yields $7x^2 - 4x - 155 = 0$. Factoring the left-hand side of this equation yields $(7x + 31)(x - 5) = 0$. Applying the zero product property to this equation yields two solutions: $x = -\frac{31}{7}$ and $x = 5$. Since x is the rectangle's width, in inches, which must be positive, the value of x is 5. Therefore, the width of the rectangle, in inches, is 5.

Q10

B

B. 3

Choice B is correct. An x -intercept of a graph in the xy -plane is a point on the graph where the value of y is 0. Substituting 0 for y in the given equation yields $0 = 2x - dx + dx + gx - d$. By the zero product property, the solutions to this equation are $x = d$, $x = -d$, $x = -g$, and $x = d$.

However, $x = d$ and $x = d$ are identical. It's given that d and g are unique positive constants. It follows that the equation $0 = 2x - dx + dx + gx - d$ has 3 unique solutions: $x = d$, $x = -d$, and $x = -g$. Thus, the graph of the given equation has 3 distinct x -intercepts.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank